

Artificial Intelligence Lab (AI-1002)

Mid Term Lab Exam

Total Time (mins): 100

Total Weightage: 25

Total Marks: 50

Total Questions: 03

Paper Type: B

Date: Wednesday, April 01st, 2026

Time: 01:40 PM to 03:20 PM

Lab Instructor(s):

Mr. Abdullah Riaz Shaikh

Mr. Abdullah Yaqoob

Student Name

Roll No

Section

Student Signature

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Attempt all the questions!

Important Instructions:

- Please submit the .ipynb file with the file name renamed as AI_Mid_Term_Lab_Exam_Your-Name_Your-Roll-No.ipynb.
- For Question 03 (initial code on Genetic Algorithm), get the code from the local submission folder by pressing Windows + R and entering \\172.16.5.43\Exam Material\Abdullah Shaikh\Q3.py.

LLO # 01: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search.

Question # 01:

[Weightage: 07 | Marks: 14 | Time: 25 minutes]

A forest management authority uses an autonomous firefighting drone to extinguish fires in different forest zones. The forest is represented as a graph of zones, each connected by forest paths. Some zones are on fire (high priority), some have minor smoke (medium priority), and some are safe (empty).

The drone has the following goals:

1. Extinguish zones on fire first (higher utility).
2. Minimize moves while navigating the forest efficiently (higher utility = fires extinguished quickly with fewer moves).

The drone can:

1. Move to an adjacent zone (cost = 1 move).
2. Extinguish fire if the zone is on fire (increase utility by 10 points per zone extinguished).
3. Explore the forest using DFS to reach burning zones efficiently.

Environment Representation

We represent the forest as a dictionary (graph), e.g.,

```
forest_graph = {  
  
    'ZoneA': ['ZoneB', 'ZoneC'],  
  
    'ZoneB': ['ZoneA', 'ZoneD'],  
  
    'ZoneC': ['ZoneA', 'ZoneE'],  
  
    'ZoneD': ['ZoneB'],  
  
    'ZoneE': ['ZoneC']  
  
}  
  
burning_zones = ['ZoneD', 'ZoneE']
```

You are required to perform the following tasks:

1. Create a UtilityBasedAgent class to track the forest environment, current zone, number of moves, and fires extinguished, and implement a utility() function as: $(10 * \text{fires extinguished}) - \text{moves}$. **[03 Marks]**
2. Implement a DFS function to find a path from the current zone to the nearest burning zone. **[07 Marks]**
3. Define agent actions so that it extinguishes fire if the current zone is burning; otherwise, it moves along the DFS path to the next burning zone while updating moves and environment state. **[02 Marks]**
4. Run a simulation until all burning zones are extinguished, printing the current zone, action taken, and utility after each step, and display the final utility at the end. **[02 Marks]**

LLO # 02: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search.

Question # 02:

[Weightage: 09 | Marks: 18 | Time: 35 minutes]

A wildfire has broken out in a forest, and an autonomous drone is tasked with locating trapped hikers and guiding them to safety. The forest area is represented as a 2D grid where each cell represents a section of the forest. The drone must navigate carefully to avoid hazards and reach the stranded hikers.

Grid Representation:

1. The environment is an $N \times M$ grid.
2. Each cell may contain:
 - **S** → Starting position of the drone
 - **G** → Goal (location of trapped hiker)
 - **.** → Clear forest path (cost = 1)
 - **#** → Dense trees (cannot pass)
 - **W** → Muddy swamp (cost = 3)
 - **F** → Fire zone (cost = 5)

The drone must find the optimal path from S to G such that the total cost is minimized.

	Col 0	Col 1	Col 2	Col 3	Col 4
Row 0	S	.	F	.	.
Row 1	W	#	W	#	.
Row 2	.	.	.	F	.
Row 3	#	.	.	.	G
Row 4	.	W	.	.	.

You are required to perform the following tasks:

1. Model the problem by defining the state space (all grid positions), initial state (S), goal state (G), allowed actions (up, down, left, right), and path cost based on terrain. **[05 Marks]**
2. Apply A* search using the evaluation function ($f(n) = g(n) + h(n)$) with Manhattan Distance as the heuristic. **[09 Marks]**
3. Show the sequence of nodes expanded during the execution of the algorithm, provide the optimal path from S to G, and Calculate the total cost of the final path. **[04 Marks]**

LLO # 04: Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance.

Question # 03:

[Weightage: 09 | Marks: 18 | Time: 40 minutes]

A delivery company must visit N cities exactly once and return to the starting city. The cities are located on a 2D map with given coordinates. The company wants to minimize the total travel distance while visiting all cities.

Design a Genetic Algorithm to find an optimal or near-optimal route that:

1. Visits each of the N cities exactly once.
2. Returns to the starting city after visiting all cities.
3. Minimizes the total travel distance of the route.

$$\text{TotalDistance} = \sum_{k=0}^{N-1} d(c_k, c_{k+1})$$

You are provided with initial code. You are required to:

1. Calculate total distance for selection. [Formula Given] **[02 Marks]**
2. Use Roulette Wheel Selection for parent selection. **[03 Marks]**
3. Use Two Point Crossover for crossover. **[04 Marks]**
4. For mutation, use Inversion Mutation. **[03 Marks]**
5. Stop when a predefined number of generations is reached, or when a route with desired minimal distance is found. **[06 Marks]**

Input:

1. Number of cities N.
2. Coordinates of each city (x_i, y_i) for i = 0 to N-1.

Output:

1. A route (ordered list of cities) showing the sequence in which to visit the cities.
2. Total distance of the route.

Input (5 cities):

City 0: (2,3)

City 1: (5,7)

City 2: (1,9)

City 3: (8,2)

City 4: (4,5)

Output:

Best Route: [4, 2, 1, 0, 3]

Total Distance: 23.87 units

Must return your paper after completion!

Best of luck!